

RESBIDET: EFFICIENT DUAL-BRANCH SMALL OBJECT DETECTION FOR UAVS UNDER RESOURCE-CONSTRAINED CONDITIONS

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ABSTRACT

Real-time Unmanned Aerial Vehicle (UAV) object detection is challenged by strict onboard resource constraints, particularly when processing small objects in high-resolution images where existing methods struggle to balance accuracy and efficiency. To address this fundamental constraint, this paper proposes a dual-branch lightweight detector (ResBiDet). The model intelligently allocates computational resources via an innovative dual-branch architecture: a lightweight, coarse-grained branch rapidly processes the full image at low resolution to locate potential object-dense regions. This in turn guides a high-resolution, fine-grained branch to focus computational power on these critical areas for precise detection. Experiments on the VisDrone2019 and UAVDT datasets demonstrate that ResBiDet achieves AP50 scores of 38.7% and 40.2%, respectively. Our approach significantly enhances small object detection accuracy while effectively reducing computational consumption, offering a high-performance solution for resource-constrained aerial applications.

Index Terms— dual-branch detection, edge computing, small object detection, lightweight model.

1. INTRODUCTION

The expanding use of Unmanned Aerial Vehicles (UAVs) in critical scenarios such as disaster relief and traffic monitoring necessitates robust real-time object detection[1],[2]. However, aerial imagery poses unique challenges—drastic scale variations, dense objects, and complex backgrounds—that impair the performance of conventional detectors[3]. The core challenge lies in the fundamental conflict between the severely limited computational resources of UAVs[4] and the high-resolution processing required to detect small targets[4],[5]. This tension between low-resource platforms and high computational demand has become a critical bottleneck for the field[2]. To address this, current methods are caught in a difficult trade-off. Downsampling high-resolution images reduces computation but loses critical features of small objects, leading to a significant drop in accuracy (Fig. 1(b)). Alternatively, image cropping strategies, whether uniform[6],[7] or guided[8],[9], are often inefficient due to resource wastage or additional overhead, hindering real-time performance[10]. This necessitates a more effective strategy that selectively processes key information, as illustrated by the foreground-preserving approach in Fig. 1(a).

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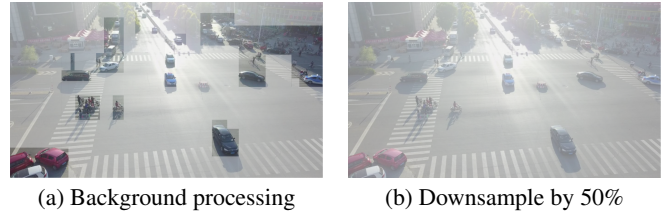


Fig. 1. Image Processing Method Comparison.

In surveying existing technologies, we noted that anchor-free models, which base detection on object center points, such as YOLC[11], provide a promising baseline for addressing aerial detection problems due to their simple architecture and good adaptability to small objects. However, the traditional “detect-crop-redetect” serial pattern is inefficient[8]. We aim to design a new architecture that intelligently allocates computational resources from the very beginning. The main contributions of this paper are as follows:

- 1) We propose a dual-branch lightweight architecture (ResBiDet), which effectively resolves the core conflict between accuracy and efficiency on resource-constrained platforms through a resource bifurcation mechanism.
- 2) We design an efficient self-guiding mechanism where a coarse-grained branch provides precise regional guidance to a fine-grained branch, achieving end-to-end coordinated detection.
- 3) We have conducted comprehensive validations on public datasets such as VisDrone2019 and UAVDT, as well as on a real-world UAV platform. The results demonstrate that ResBiDet significantly improves the detection accuracy of small objects while effectively reducing resource consumption, showcasing superior overall performance.

2. RELATED WORKS

To address the severe challenges of small object detection in UAV aerial imagery[12], the academic community has developed various technical approaches. This section provides a critical review of existing work from two main perspectives—accuracy-first and efficiency-first strategies—to clarify the innovative positioning of ResBiDet.

2.1. Accuracy-First Strategies and Their Efficiency Bottlenecks

Two primary strategies exist for enhancing the detection accuracy of small objects: feature enhancement[13], such as in FFCA-YOLO[14], and region-focusing, like image slicing[6],[7]. Although these methods have achieved significant progress in accuracy, they

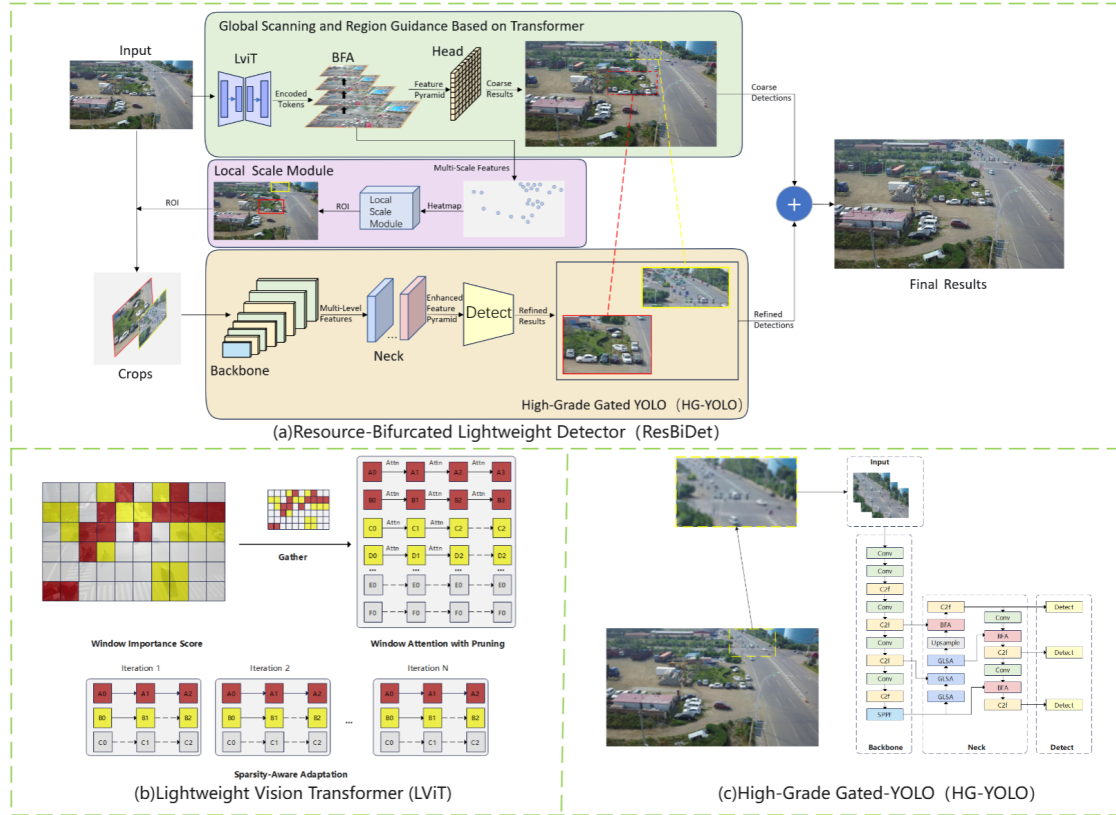


Fig. 2. The architecture of the proposed ResBiDet model. The framework features a dual-branch design where the upper branch utilizes a Lightweight Vision Transformer (LViT) to perform an efficient global scan via a sparse attention mechanism and a Bidirectional Feature Aggregation (BFA) module to efficiently fuse multi-scale features. The resulting heatmap guides the Local Scaling Module (LSM) to crop small-object-dense regions. The lower branch then employs the High-Grade Gated YOLO (HG-YOLO) detector to conduct high-precision detection on these cropped regions. The final result is produced by fusing the outputs from both branches.

generally face efficiency bottlenecks on resource-constrained platforms such as UAVs. The inefficiency stems from their core designs. Global feature enhancement performs indiscriminate computations across the entire image, leading to a substantial waste of resources on background regions[15],[16]. On the other hand, region-focusing strategies introduce new computational overhead. Whether implemented as a separate preprocessing step or requiring an additional network for guidance, their non-integrated nature makes it difficult to achieve true end-to-end efficient detection.

2.2. Efficiency-First Strategies and Their Accuracy Compromises

To meet the deployment requirements of edge platforms, another class of methods concentrates on improving model efficiency. This is achieved primarily through two avenues: designing lightweight network architectures, such as EfficientDet[16] and YOLO variants[17],[18] or introducing efficient computation paradigms, like EfficientViT[19]. However, in their pursuit of global lightweighting, these methods often compromise by sacrificing network depth, resolution, or model capacity[20]. This "one-size-fits-all" compression strategy, while reducing the overall computational burden, may also impair the feature representation capabilities necessary for identifying critical small objects[21]. This leads to a dilemma where existing works are trapped between "high

accuracy-high cost" and "high efficiency-low accuracy." Therefore, designing a new paradigm that can intelligently allocate resources to better balance accuracy and efficiency is a key breakthrough urgently needed in this field.

3. PROPOSED METHOD

To effectively address the challenge of small object detection for UAVs in resource-constrained environments, we design a dual-branch lightweight detector named ResBiDet. As shown in Fig. 2, the overall design of ResBiDet follows a "coarse-to-fine" detection paradigm[22],[23]. By intelligently bifurcating the computational tasks into two collaboratively working branches, it achieves an optimal balance between global perception and local detail enhancement. The architecture mainly consists of three core components: (1) Global scanning and region guidance based on sparse attention (2) High-Grade Gated YOLO (HG-YOLO) (3) dual-branch collaborative training with hierarchical inference fusion.

3.1. Global Scanning and Region Guidance Based on Transformer

The global branch (upper path in Fig. 2(a)) is engineered to efficiently process the entire high-resolution image. The original input is first passed through our proposed Lightweight Vision Transformer

(LViT) backbone and a Bidirectional Feature Aggregation (BFA) module for efficient multi-scale feature extraction and fusion [24]. To further enhance the efficiency of the feature pyramid, we introduce the BFA module, which improves upon the classic BiFPN [16] by incorporating a lightweight attention mechanism to dynamically adjust the importance of features at different scales and by optimizing cross-scale connection paths, enabling more effective multi-scale fusion at a lower computational cost.

Subsequently, the fused features are split into two paths: one feeds a detection head to generate initial coarse predictions, while the other produces a localization heatmap. This heatmap is aggregated into object-dense regions via threshold-based window selection and serves as region-level guidance for the Local Scale Module (LSM), a density-heatmap-guided cropping module that identifies and crops small-object-dense patches from the original high-resolution image for subsequent fine-grained local detection.

The core of this branch is the LViT backbone. LViT builds on the core concept of SparseViT [15], which, as shown in Fig. 2(b), leverages a window-level token pruning mechanism to focus computation on informative regions. We further optimize its importance scoring and sparse attention computation to more accurately identify potential object-dense regions while maintaining high efficiency. Unlike traditional CNNs or ViTs [25], LViT can bypass background computations and concentrate resources on information-rich areas, providing a natural efficiency advantage for aerial image processing.

In summary, through its "sparse-fuse-distribute" workflow, this branch not only performs preliminary global detection but, more crucially, provides precise guidance for the entire architecture, enabling intelligent resource allocation.

3.2. High-Grade Gated YOLO

The fine-grained branch (lower path in Fig. 2(a)) is the core of ResBiDet for achieving high-precision small object detection[14],[11]. Its task is to receive the high-value image crops filtered by the global branch and perform refined object detection on these local high-resolution regions. The input of this branch comes from the Local Scaling Module (LSM), which determines the cropping regions based on the localization heatmap H generated by the global branch. Specifically, LSM converts the heatmap into a binary region mask M through a predefined density threshold:

$$M(W_k) = \begin{cases} 1 & \text{if } \sum_{(i,j) \in W_k} H(i,j) > T_{density} \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

where $M(W_k)$ denotes the decision for the k -th window. Subsequently, by connecting all contiguous regions with value 1, we generate the final cropping region proposals.

To achieve the optimal balance between accuracy and efficiency within this branch, we designed a lightweight yet high-performance detector named HG-YOLO (High-Grade Gated YOLO). As illustrated in Fig. 2(c), HG-YOLO is based on the compact architecture of YOLOv8n, and we replace the original feature pyramid network (FPN) with our Bidirectional Feature Aggregation (BFA) module. BFA introduces learnable weights to dynamically adjust the importance of features at different scales, achieving more comprehensive and efficient multi-scale feature fusion. The weighted fusion process can be formulated as:

$$P_{out} = Conv \left(\frac{\sum_i w_i \cdot I_i}{\sum_i w_i + \epsilon} \right) \quad (2)$$

where I_i denotes the input feature maps from different levels, and ϵ is a small constant added for numerical stability. To further enhance

the model's perception ability for tiny objects, we embed a Global-Local Sparse Attention (GLSA) module after the feature maps F_p at each detection level ($P3, P4, P5$). This module strengthens the model's focus on fine-grained local features while maintaining sparse activation, thereby effectively suppressing interference from complex backgrounds[15]. Its operation can be formalized as:

$$F_{lp} = GLSA(F_p) \quad (3)$$

where F_{lp} represents the attention-enhanced feature map. By combining the powerful fusion capability of BFA with the precise attention mechanism of GLSA, HG-YOLO is able to perform robust feature extraction and analysis on the input high-resolution crops, generating highly accurate refined results.

3.3. Dual-Branch Collaborative Training and Hierarchical Inference Fusion

The primary advantage of ResBiDet stems from the synergistic interplay of its dual branches, which is evident in both the joint optimization during training and the efficient fusion during inference. To facilitate this collaborative training, we developed a multi-component loss function consisting of three terms: a global branch detection loss L_G , a local branch detection loss L_R , and a cross-branch consistency loss L_{cons} . The global branch loss L_G adopts a hybrid strategy, combining focal loss, L_1 loss, and a geometry loss based on the Gaussian Wasserstein distance. Among these, the geometric distance loss L_{dist}^G uses the 2nd order Wasserstein distance to uniformly constrain the center point offset and the width-height differences between the predicted and ground-truth boxes. Its core is based on the calculation of W^2 :

$$W^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2 + \frac{(w_1 - w_2)^2 + (h_1 - h_2)^2}{4} \quad (4)$$

The final geometric distance loss function is expressed as:

$$L_{dist}^G = 1 - \frac{1}{\tau + \ln(1 + W^2)}, \quad \tau \geq 1 \quad (5)$$

Likewise, a similar loss function L_R is adopted to supervise the local branch. The model's total loss function, L_{total} , is ultimately the weighted sum of these individual losses:

$$L_{total} = L_G + \alpha L_R + \beta L_{cons} \quad (6)$$

By performing end-to-end optimization with this collaborative loss function, the two branches of ResBiDet learn division of labor and cooperation during training, laying a solid foundation for efficient inference fusion.

At the inference stage, ResBiDet employs a "Replace-and-Merge" strategy. First, it preserves the coarse detections (Coarse Results) from the global branch. Then, the refined detections (Refined Results) from the local branch directly replace all predictions within their corresponding spatial areas in the coarse output. To conclude, a standard Non-Maximum Suppression (NMS) operation[26], denoted by the $\oplus$ symbol in Fig.2(a), is performed on the fused results to yield the final, globally consistent, and high-precision detections (Final Results).

4. EXPERIMENT RESULTS AND ANALYSIS

To comprehensively evaluate the proposed ResBiDet model's performance under resource-constrained environments, we conducted a series of rigorous experiments, including comparisons with various state-of-the-art detectors and ablation studies on the model's key components.

Table 1. Comparison of UAV Object Detection Models on Visdrone and UAVDT

Model	Params (M)	AP50 (Visdrone)	AP50 (UAVDT)	FPS	AP50/Params (Visdrone)	AP50/Params (UAVDT)	Derivation
RT-DETR[27]	86.0	49.3	53.2	68	0.573	0.619	CVPR 2024
QueryDet[8]	42.2	47.1	45.7	25	1.116	1.083	CVPR 2023
DEIM[28]	32.6	39.9	42.7	33	1.224	1.310	CVPR 2025
Faster R-CNN[22]	42.3	18.4	19.7	12	0.435	0.466	NeurIPS 2015
DMNet[29]	34.0	28.2	29.4	47	1.124	0.865	CVPR 2020
FFCA-YOLO[14]	10.3	34.2	36.5	63	3.320	3.544	TGRS 2024
YOLOv8n	6.0	31.5	32.3	73	5.250	5.383	Ultraliyolovics 2023
YOLC(Baseline)[11]	47.9	30.9	32.1	27	0.645	0.670	TITS 2024
ResBiDet	25.1	38.7	40.2	81	1.542	1.602	–

4.1. Experimental Setup

Our experiments were conducted on two challenging public aerial image datasets, *VisDrone2019* and *UAVDT*, which feature complex scenes with numerous small and densely distributed objects, closely reflecting real-world UAV scenarios. To enrich small-object samples, we adopted a density-map-guided cropping strategy using a pre-trained MCNN to extract high-density patches from high-resolution images for training. In addition, real-world experiments were carried out on a physical UAV platform equipped with an NVIDIA Jetson Orin Nano with 8 GB GPU memory, and efficiency was evaluated by jointly considering AP50, FPS, and the performance/parameter ratio rather than relying solely on FPS, providing a more comprehensive assessment of the balance between accuracy and efficiency.

Table 2. Ablation Study of Different Components

Model	LVIT	BFAa	GLSA	BFAb	AP50 (Visdrone)	AP50 (UAVDT)	Params
YOLC(Baseline)[11]	×	×	×	×	30.9	32.1	47.9
A	✓	×	×	×	33.6	35.0	31.8
B	×	✓	×	×	32.8	34.0	37.5
C	×	×	✓	×	32.4	33.9	42.8
D	×	×	×	✓	33.5	35.2	36.3
ResBiDet	✓	✓	✓	✓	38.7	40.2	25.1

4.2. Comparison with State-of-the-Art Methods

In comparative experiments, ResBiDet demonstrated a superior balance of accuracy and efficiency while maintaining a lightweight profile. As shown in Table 1, when compared to high-accuracy models like QueryDet[8], RT-DETR[27], and DEIM[28], ResBiDet achieved highly competitive performance on the AP50 metric, while its inference speed of 81 FPS far surpassed their 25–68 FPS, highlighting its excellent real-time capabilities. Against classic models such as Faster R-CNN[22] and DMNet[29], ResBiDet achieved substantial improvements across all accuracy metrics while operating with greater efficiency. In comparisons with advanced lightweight models like YOLOv8n and FFCA-YOLO[14], ResBiDet, with a comparable or slightly larger parameter count, achieved significant AP50 gains of 7.2% and 7.9% (vs. YOLOv8n) on the VisDrone and UAVDT datasets, respectively, proving the superiority of its architectural design. The advantage over the baseline model, YOLC, was particularly pronounced: not only was the parameter count reduced by 47.6% (from 47.9M to 25.1M), but the AP50 also increased by 7.8% and 8.1% on the two datasets. Overall, ResBiDet’s unique resource-bifurcation design achieves an exceptional accuracy-efficiency trade-off. Its “accuracy-to-parameter ratio”

(AP50/Params) reached 1.542 and 1.602 on VisDrone and UAVDT, respectively, outperforming heavyweight competitors like QueryDet and RT-DETR, and demonstrating a remarkable balance between model size and performance.

4.3. Ablation Study

To validate the effectiveness of each core component within ResBiDet, a series of detailed ablation studies were conducted with YOLC as the baseline model. The results are presented in Table 2. The baseline model, without any of the proposed components, achieves an AP50 of 30.9% on VisDrone and 32.1% on UAVDT with 47.9M parameters.

First, the performance of the LVIT backbone was evaluated. By solely replacing the baseline’s backbone with LVIT, AP50 scores increased to 33.6% and 35.0% on VisDrone and UAVDT, respectively, while the parameter count was significantly reduced to 31.8M. Subsequently, modules in the neck network like BFAa, GLSA, and BFAb were independently tested, and all achieved performance gains over the baseline, which validates their effectiveness in feature fusion and attention enhancement.

Ultimately, the complete ResBiDet model, which integrates all components, achieved the best overall performance. Its AP50 reached 38.7% on VisDrone and 40.2% on UAVDT with an optimized parameter count of 25.1M. Compared to the YOLC baseline, ResBiDet improved accuracy by 7.8 percentage points on VisDrone while cutting the parameter count by 47.6%. The ablation study confirms the modules’ synergistic advantages and indispensable contributions to the model’s high accuracy and efficiency.

5. CONCLUSIONS

In this paper, we propose ResBiDet, a dual-branch lightweight detector that addresses the key challenge of balancing small-object detection accuracy and efficiency under UAV resource constraints. With a novel coarse-to-fine dual-branch architecture, the model enables guided and efficient allocation of computational resources. ResBiDet achieves 38.7% and 40.2% AP50 on VisDrone2019 and UAVDT, respectively, and demonstrates strong energy efficiency in real onboard experiments, confirming its balanced performance in accuracy, efficiency, and resource usage.

Future work will explore more dynamic resource scheduling and extend the framework to multi-modal fusion, making ResBiDet a promising paradigm for efficient edge-device visual perception.

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